

WEEK 2 · FINAL EXAM · ALL NINE COURSES

Week 2 Final Exam

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **17 of 24** right to pass and earn your Week 2 diploma.

1. Which of these sentences from Monday's Biology is a BIG IDEA rather than a detail?

☐

A. The umbilical cord has three vessels: two arteries carrying blood out and one vein bringing it back.

☐

B. William Harvey was the personal physician to King Charles I of England in the 1620s.

☐

C. A newborn's heart is about the size of a walnut, and an adult's is about the size of a fist.

☐

D. The heart is two pumps: the right side sends blood to the lungs, the left side sends it to the body.

2. Blood came up in Chemistry, Biology, Physics, Genetics, and Calculus this week. Which statement is TRUE?

☐

A. Its pH is set by the bones, its platelets carry the oxygen, and a 1 kg preemie has about 5 liters of it.

☐

B. Its pH must stay near 5.5, its plasma carries the oxygen, and a 1 kg preemie has about 800 mL of it.

☐

C. Its pH must stay near 7.4, its red cells change color with oxygen, and a 1 kg preemie has only about 80 mL of it.

☐

D. Its pH must stay near 7.4, its white cells carry the oxygen, and a 1 kg preemie has about 8 mL of it.

3. A blood gas shows pH 7.18 and a CO₂ far above normal. What's the story?

☐

A. The baby is breathing far too much and blowing off acid – turn the ventilator settings down.

☐

B. Carbonic acid is piling up because the lungs aren't clearing CO₂ – give more breathing help.

☐

C. The cells are starving for oxygen and making lactic acid – start antibiotics at once.

☐

D. The kidneys have lost all their bicarbonate – a kidney problem; give bicarbonate right away.

4. Lemon juice is about pH 2 and black coffee about pH 5. How many times more acidic is the lemon juice?

- ☐ A. About 300 times
- ☐ B. About 30 times
- ☐ C. About 3 times
- ☐ D. About 1,000 times

5. Where does the LEFT side of the heart pump blood?

- ☐ A. Out to the whole body – brain, gut, kidneys, fingertips.
- ☐ B. To the placenta, through the vein in the umbilical cord.
- ☐ C. To the lungs, to drop off carbon dioxide and pick up oxygen.
- ☐ D. Back to the right side, through the ductus arteriosus.

6. A preemie has a PDA. What is happening?

- ☐ A. The umbilical vein never closed, so blood keeps flowing out through the belly button.
- ☐ B. A valve between the atria stayed shut, so blood can't get from the right heart to the lungs at all.
- ☐ C. A fetal bypass pipe near the heart stayed open, so blood leaks back into the lungs and the body gets less.
- ☐ D. The heart's electrical spark is too slow, so the baby's heart rate keeps dropping.

7. Why does the pulse oximeter use both red and infrared light?

- ☐ A. Oxygen-loaded and empty hemoglobin absorb the two colors differently, so comparing them gives the percentage.
- ☐ B. Two colors together are bright enough to shine all the way through a baby's foot.
- ☐ C. Red light finds the arteries and infrared light finds the veins, and the machine adds them.
- ☐ D. Red light measures the heart rate and infrared light measures how warm the foot is.

8. What does blue light do to bilirubin?

- ☐ A. It bleaches the yellow color out of the skin so the bilirubin underneath can no longer be seen.
- ☐ B. Bilirubin absorbs it, and the energy twists the molecule into a shape the body can flush out.
- ☐ C. It warms the blood flowing under the skin so that the liver's slow enzyme starts working faster.
- ☐ D. It breaks apart the red blood cells that make bilirubin, so that no more of it is produced.

9. A pump runs at 2.5 mL per hour for 8 hours. How much did the baby get?

- ☐ A. 2.5 mL
- ☐ B. 20 mL
- ☐ C. 32 mL
- ☐ D. 10.5 mL

10. On a graph with the pump rate as height and time along the bottom, the total delivered is...

- ☐ A. the steepness of the rate line.
- ☐ B. the area under the rate line.
- ☐ C. the number of hours along the bottom.
- ☐ D. the height of the rate line at the end.

11. Seven babies weigh 900, 1,300, 820, 950, 2,400, 880, and 1,000 grams. What is the median?

- ☐ A. 1,179 g
- ☐ B. 900 g
- ☐ C. 950 g
- ☐ D. 1,000 g

12. A baby was on the 50th percentile line a month ago and is on the 10th today. A doctor would say...

- ☐ A. she is falling off her curve — an alarm, even though today's weight is still 'normal.'
- ☐ B. she is at a healthy weight, because the 10th percentile is still on the chart.
- ☐ C. she has moved up in line, because 10 is a lower number than 50.
- ☐ D. nothing, because a percentile is a grade and she hasn't failed yet.

13. In PKU, what is broken and what happens because of it?

- ☐ A. The enzyme that breaks down phenylalanine, so phenylalanine piles up and harms the brain.
- ☐ B. The thyroid gland, so a growth hormone runs short and the baby grows slowly.
- ☐ C. The liver's bilirubin enzyme, so the baby turns yellow in the first week.
- ☐ D. The hemoglobin recipe, so red blood cells bend into sickle shapes and jam.

14. Why is every baby screened for diseases as rare as 1 in 12,000?

- ☐ A. Because they hide at birth, waiting causes harm, early treatment works, and the test is cheap and safe.
- ☐ B. Because hospitals are required to collect a blood sample from every baby for its records.
- ☐ C. Because the heel prick also gives the baby its first vaccination against them.
- ☐ D. Because most babies turn out to have at least one of the conditions on the card.

15. What did Harlow's baby monkeys teach psychologists?

- ☐ A. Baby monkeys can't tell a real mother from a pretend one made of cloth, so they clung to both.
- ☐ B. Comfort matters more than food – they clung to the cloth mother and only visited the wire one to eat.
- ☐ C. Monkeys raised with two pretend mothers grow up healthier than monkeys raised with just one.
- ☐ D. Food matters more than comfort – they stayed on the wire mother that held the milk bottle all day.

16. In the still-face experiment, a mother goes blank for two minutes. What does this show about babies?

- ☐ A. They don't notice faces at all until about six months old, so nothing much happens for two minutes.
- ☐ B. They prefer a calm, still face to a moving one, and settle down quickly once the mother stops.
- ☐ C. They forget a face within a few seconds, so they treat their own mother as a complete stranger.
- ☐ D. They expect serve-and-return and work hard to restart it – then recover fast when the face returns.

17. Country A has 200 preterm babies out of 2,000 births. Country B has 300 preterm babies out of 6,000 births. Which is TRUE?

- ☐ A. B has both more preterm babies and the higher rate, because 300 is bigger than 200.
- ☐ B. They have the same rate, because both countries are counted per thousand.
- ☐ C. B has more preterm babies, but A has the higher preterm rate (10 per 100 versus 5 per 100).
- ☐ D. A has more preterm babies and the higher rate, because 200 is more than 5 per 100.

18. In the world's poorest countries, about half of babies born at 32 weeks or earlier die. Why, mostly?

- ☐ A. Their parents choose to care for them at home instead of in hospitals, for religious reasons.
- ☐ B. The diseases there are different from the ones that rich-country NICUs know how to treat well.
- ☐ C. Babies there are born weaker than babies elsewhere, so the same care would not have saved them.
- ☐ D. They don't get basic care – warmth, feeding help, and treatment for infections and breathing trouble.

19. Flashback to Week 1 Physics: why is blood pressure measured in millimeters of mercury?

- ☐ A. Doctors chose mercury because it was the fastest planet, and blood is the fastest-moving fluid.
- ☐ B. Mercury is the silvery metal that the first blood-pressure cuffs were made from in the 1890s.
- ☐ C. The first pressure meters were literally a column of mercury: the push is how high it lifted the liquid.
- ☐ D. Blood is about as heavy as mercury, so the two liquids balance each other inside the meter.

20. Flashback to Week 1 Statistics: a baby's Apgar score is 5 at one minute. What does that mean?

- ☐ A. The baby may need some help right now – rubbing, warming, perhaps a little oxygen.
- ☐ B. The score was taken too early to mean anything and must be thrown out and redone.
- ☐ C. The baby will have lifelong problems, because 5 out of 10 is a failing score.
- ☐ D. The baby is doing well and needs only routine newborn care from the nurses.

21. Flashback to Week 1 Genetics: what is a gene?

- ☐ A. One of the four chemical letters, A, T, C, or G, that DNA is written in.
- ☐ B. One recipe in the DNA instruction book – the instructions for building one protein.
- ☐ C. One of the 46 volumes of the instruction book, half from each parent.
- ☐ D. A typo in the instruction book that changes how a protein is built.

22. Flashback: what does the isolette's skin probe do?

- ☐ A. It measures the baby's oxygen level by shining red and infrared light through the skin.
- ☐ B. It checks how much water is evaporating from the baby's thin skin and adjusts the humidity.
- ☐ C. It reports the baby's skin temperature so the isolette can warm or cool its air to hit the target.
- ☐ D. It counts the baby's heartbeats from the tiny electricity that leaks out to the skin.

23. Flashback: why is caffeine one of the most important medicines in a NICU?

- ☐ A. It makes a preemie's blood a little more acidic so the lungs work harder.
- ☐ B. It helps a preemie's liver clear bilirubin faster, so the skin turns less yellow.
- ☐ C. It speeds up a preemie's heart so that more blood reaches the growing brain.
- ☐ D. It keeps a preemie's immature breathing center awake, preventing apnea spells.

24. Flashback: why can't a baby born at 26 weeks drink from a bottle?

- ☐ A. Its stomach can't digest milk of any kind until about 34 weeks, so it gets only sugar water in an IV.
- ☐ B. Its teeth haven't come in yet, so it can't hold a bottle's nipple in place long enough to drink.
- ☐ C. Its mouth is far too small to fit around any bottle at that age, so it must be fed through a tube instead.
- ☐ D. The suck-swallow-breathe rhythm is a brain skill wired at about 32 to 34 weeks; milk would go into the lungs.

Score: _____ / 24

Badge earned? YES / NOT YET (need 17)